Nova School of Business and Economics
Calculus II 2025 – 2026 S1
Áurea Quintino
João Fanha
Pamela Pacciani

Midterm Test 2

- **Date:** November 08, 2025
- **Duration:** 1 hour
- **Instructions: 1:** This test has 10 questions. 2: All questions are of multiple choice and **each question admits a single correct answer**. 3: **Each correct answer, wrong answer and blank answer submitted is classified with, respectively, 1, - 0.15 and 0 points.** 4: No written support or calculators are allowed. 5: **You may freely navigate** through the questions, choosing the answer to each one at any time, before you submit your answers. 6: On the right side of each page and at the end of the last page, you may click on Finish attempt when you are ready to submit all your answers. However, when you do this, you still do not submit your answers. You may click on Return to attempt and continue changing your answers, or on Submit all and finish and submit them. 7: In order to have your answers submitted, you need to **click on Submit all and finish** and, after that, **ask the invigilator to introduce the exit password.** **8**: If time runs out before you submit your answers, they will be automatically saved as they are. Before you leave the room, do not forget to **ask the invigilator to introduce the exit password.**

  All the best on your test!

Nr:

1.

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = -x^2 - y^2 + y^2 x.$$

Which of the following propositions is **true**?

- ○ The stationary points of f are $(0, 0)$ and $\left(1, \sqrt{2}\right)$.
- ○ $(0, 0)$ is a strict local maximizer of f, and f does not admit minimizers.
- ○ $(0, 0)$ and $\left(1, \sqrt{2}\right)$ are saddle points of f.
- ○ f does not admit extremum points.

2.

Consider a function $f : \mathbb{R} \to \mathbb{R}$ such that:

- f is C^2
- the Taylor polynomial of degree 2 of f around 2 is $P_2^2(x) = 1 - 3(x-2) + 5(x-2)^2$.

Which of the following options is **necessarily true**?

- For f^{-1} a local inverse of f at 2, $(f^{-1})'(2) = (f'(2))^{-1}$.
- For f^{-1} a local inverse of f at 2, $(f^{-1})'(1) = \frac{-1}{3}$.
- For f^{-1} a local inverse of f at 2, $(f^{-1})'(1) = (f'(1))^{-1}$.
- None of the remaining options is necessarily true.

3.

Consider two functions, $f : \mathbb{R} \to \mathbb{R}$ and $F : \mathbb{R}^2 \to \mathbb{R}$, such that:

- $f'(0) = -1$
- f is C^1
- f only vanishes at 0
- F is defined by $F(x, y) = \int_{x-y}^{1} f(t)\, dt$.

Which of the following propositions is **FALSE**?

- $H_F(x, x)$ is positive semi-definite, for all $x \in \mathbb{R}$.
- One of the remaining options is a false proposition.
- Any point on the straight line with equation $y = x$ is a stationary point of F.
- $H_F(x, x) = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, for all $x \in \mathbb{R}$.

4.

Let $q : \mathbb{R}^3 \to \mathbb{R}$ be the quadratic form defined by

$$q(x, y, z) = -2x^2 - y^2 - 2z^2 + 4xz.$$

What is the classification of q regarding sign?

- q is negative definite.
- q is negative semi-definite, but it is not negative definite.
- None of the remaining options.
- q is indefinite.

5.

Consider two functions, $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R}^2 \to \mathbb{R}^2$, such that:

- f is C^1
- $f(1,1) = 2$
- $\nabla_f (1,1) = (1,2)$
- $g(x,y) = (f(x+1, -y+1), f(y+1, x+1)), \forall (x,y)$.

Let g^{-1} be a local inverse of g at $(0,0)$.

Which of the following propositions is **necessarily true**?

- $5J_{g^{-1}}(2,2) = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$.
- None of the remaining options is necessarily true.
- The matrix $J_g(0,0)$ is not invertible.
- $J_{g^{-1}}(0,0) = \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{-2}{5} & \frac{1}{5} \end{bmatrix}$.

6.

Let A be a 3×3 symmetric matrix with determinant equal to -6 and admitting 2 as an eigenvalue.

Let q be the quadratic form whose associated symmetric matrix is A.

What is the classification of q regarding sign?

- None of the remaining options.
- q is negative semi-definite, but it is not negative definite.
- q is negative definite.
- q is indefinite.

7.

Consider the following equation:

$\mathbb{E} : 2\ln(x+y) + y^2 = 4$

In a neighbourhood of $(x,y) = (-1,2)$, $\mathbb{E}$ implicitly defines a function f such that $x = f(y)$.

Which of the following propositions is **necessarily true**?

- $f'(-1) = 3$
- $f'(2) = -3$
- $f'(-1) = \frac{1}{3}$
- $f'(2) = \frac{1}{12}$

8.

Consider the equation

$\mathbb{E} : y + e^y + xz = 1.$

In a neighbourhood of $(x, y, z) = (0, 0, 1)$, $\mathbb{E}$ implicitly defines a function f such that $y = f(x, z)$.

Which of the following propositions is **necessarily true**?

- f is not C^∞.
- The Taylor polynomial of degree 1 of f around $(0, 1)$ is $P(x, z) = -\frac{x}{2}$.
- $(0, 1)$ is a stationary point of f.
- $f(0, 0) = 1$.

9.

Consider the funtion $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$f(x, y) = x^2 + y^2 - 2x - 6y + 14.$

Which of the following propositions is **FALSE**?

- $(1, 3)$ is a strict global minimizer of f.
- 2 is the global minimum of f.
- f is strictly convex.
- $(1, 3)$ is the only stationary point of f.

10.

Consider the equation

$\mathbb{E} : y + e^y + xz = 1.$

In a neighbourhood of $(x, y, z) = (0, 0, 1)$, $\mathbb{E}$ implicitly defines a function f such that $y = f(x, z)$.

What is the value of $f''_{xx}(0, 1)$?

- 4^{-1}
- -8^{-1}
- 2^{-1}
- 0

Solution

1. $(0,0)$ is a strict local maximizer of f, and f does not admit minimizers.

2. For f^{-1} a local inverse of f at 2, $(f^{-1})'(1) = \frac{-1}{3}$.

3. One of the remaining options is a false proposition.

4. q is negative semi-definite, but it is not negative definite.

5. $5\, J_{g^{-1}}(2,2) = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$.

6. q is indefinite.

7. $f'(2) = -3$

8. The Taylor polynomial of degree 1 of f around $(0,1)$ is $P(x,z) = -\frac{x}{2}$.

9. 2 is the global minimum of f.

10. -8^{-1}